

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA02

COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO1: Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:100

Annexure A

PSEUDO CODE WRITING

Code of Conduct:

I HARSHAN V [192412213] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

V. Harshan

Signature of the student:

Annexure A

PSEUDO CODE WRITING

Q1. Given pseudocode:
#define DEBUG 0

```
IF DEBUG
  PRINT intermediate_value
END IF
```

- Identify error in directive usage
- Correct the logic

Q2. #define PI _____

```
DECLARE radius AS float
DECLARE area AS float
```

```
INPUT radius
```

```
area = _____
```

```
PRINT area
```

Fill:

- Constant
- Formula

Q3. #define RATE 1.5

```
DECLARE x AS int = 10
DECLARE result AS float
```

```
result = x * RATE
```

```
PRINT result
```

Questions:

- Output?
- Why is the result float?

Q4. #define MAX 1000

DECLARE temp AS int
INPUT temp

IF temp > MAX
 PRINT "Valid"
ELSE
 PRINT "Invalid"

- Identify logical error
- Suggest correct condition

Q5. Given unordered pseudocode steps:

- DEFINE conversion rate
- PRINT result
- INPUT amount
- CALCULATE converted value

Arrange in correct order

Q6. #define MIN 10
#define MAX 50

DECLARE value AS int
INPUT value

_____ (value >= MIN AND value <= MAX)
PRINT "Within range"

_____ PRINT "Out of range"

Fill missing control structures

Q7. A billing system includes optional features such as discount calculation and tax computation. Depending on the business requirement, these features may be enabled or disabled at compile time. The program should calculate the total bill amount based on input price and quantity. If discount and tax features are enabled, they should be applied using predefined constants; otherwise, they should be skipped. The system must ensure that floating-point precision is maintained during calculations.

Write pseudocode using preprocessor directives to enable/disable features, define constants for discount and tax rates, and use appropriate data types.

Q8. #define TAX 5%

DECLARE price AS int
DECLARE total AS int

total = price + (price * TAX)

PRINT total

- Identify errors
- Correct data types and constant definition

9. A temperature conversion program supports multiple modes such as Celsius to Fahrenheit and Fahrenheit to Celsius. The mode of operation should be selected at compile time using preprocessor directives. The program should use appropriate floating-point data types to ensure accuracy and define conversion formulas using symbolic constants wherever applicable. The final output should be displayed with two decimal precision.

Write pseudocode using preprocessor directives for mode selection, constants for conversion formulas, and proper data types.

10. #define SIZE 5

DECLARE arr[SIZE]
INPUT elements

- Extend pseudocode to:
 - Find sum
 - Compute average

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Error: DEBUG

Correct Logic:

```
#define DEBUG 1
```

```
IF DEBUG THEN
```

```
    PRINT intermediate-value
```

```
END IF
```

2.

```
#define PI 3.14
```



```
DECLARE radius AS float
```



```
DECLARE area AS float
```



```
INPUT radius
```



```
area = PI * radius ** 2
```



```
PRINT area
```

3. Output: 15.0

- * RATE is 1.5, Which is a Floating point Constant
- * Multiplication with 1.5 is Floating Point Result.
- * Result is stored as a float.

4. Logical Error

If MAX=1000, a value greater than 1000 should generally be Considered Invalid, not Valid

Correct Condition:

```
IF temp <= Max
```

```
    PRINT "VALID"
```

```
ELSE
```

```
    PRINT "Invalid"
```

```
END IF
```

5.

Correct Order

1. Define Conversion rate
2. INPUT amount
3. Calculate Converted Value
4. PRINT result.

6.

define MIN 10

define MAX 50

DECLARE Value AS int

INPUT Value

IF (Value >= MIN AND VALUE <= MAX)

PRINT "Within Range"

ELSE

PRINT "Out of Range"

END IF

7.

Pseudocode

#define DISCOUNT_ENABLED (1) if discount is enabled

#define TAX_ENABLED (1) if tax is enabled

#define DISCOUNT_RATE 0.10

#define TAX_RATE 0.05

DECLARE price AS float

DECLARE quantity AS int

DECLARE total AS float

Input price

Input quantity

total = price * quantity

IF DISCOUNT_ENABLED Then

total = total - (total * DISCOUNT_RATE)

ENDIF

IF TAX_ENABLED Then

total = total + (total * TAX_RATE)

End if

Print "Total Bill =", total

Features Used

- * Preprocessor Directives: #define
- * Constants: DISCOUNT-RATE, TAX-RATE
- * Compile time feature Section
 - * DISCOUNT-Enabled
 - * TAX-Enabled
- * Appropriate DataTypes
 - * float for price & Total
 - * int for quantity

8.

Errors:

1. #define TAX 5% is invalid.
2. TAX should be written as a decimal (0.05)
3. price should be float.
4. total should be float.

Correct Pseudocode:

```
#define TAX 0.05
DECLARE price AS float
DECLARE total AS float
INPUT price
total = price + (price * TAX)
PRINT total
```

9.

Pseudocode

```
#define C-To-F1
#define F-To-C0
#define Factor 1.8
#define OFFSET
DECLARE temperature AS float
DECLARE result AS float
INPUT temperature
```


IF C-TO-F Then

result = (temperature * Factor) + OFFSET

END IF

IF F-TO-C Then

result = (temperature - OFFSET) / Factor

END IF

PRINT RESULT TO 2 DECIMAL PLACES

10. Pseudocode

define size 5

DECLARE arr[SIZE] AS int

DECLARE sum AS int = 0

DECLARE average AS float

DECLARE i AS int

For i = 0 TO size - 1

INPUT arr[i]

sum = sum + arr[i]

end For

average = sum / size

Print "Sum =", sum

Print "Average =", average